

Questions with ★ may be more challenging.

1 Constructions

Question 1. Let $G = (V, E)$ be an undirected graph. For $F \subseteq \binom{V}{2}$, let $G \triangle F$ denote the edge modification of G by F , i.e. for every edge $uv \in F$, if $uv \in E$ then it is removed and if not then it is added. Let $E_V = \binom{V}{2}$ for readability.

Show that the following are linear delta-matroids.

1. The delta-matroid $D = (E_V, \mathcal{F})$ where a set $F \subseteq E_V$ is feasible if and only if the edge modification $G \triangle F$ is connected.
2. The same as the last part, except that we also require that $G \triangle F$ has precisely k edges (for some $k \in \mathbb{N}$).

2 “Symmetric” delta-matroids

Question 2. Define a symmetric set system (warning: non-standard term!) as follows. Let V be a ground set of size n and $L \subseteq \{0, \dots, n\}$. Define $D_L = (V, \mathcal{F})$ where $\mathcal{F} = \{S \subseteq V \mid |S| \in L\}$. For example, a “symmetric matroid” definable this way would simply be a uniform matroid (in either basis or independent set form).

1. For which sets L is D_L a delta-matroid?
2. Among these, which are linear delta-matroids?
3. ★ For which L is D_L a projected linear delta-matroid?

For items 2 and 3, make sure that you provide both upper and lower bounds (i.e. show that every positive case you claim can actually be constructed).

Remark: There is a notion of linear delta-matroids represented over symmetric matrices, which has no relation to the above definition.

3 Strong exchange axioms

A delta-matroid $D = (V, \mathcal{F})$ is *strong* if it satisfies the following variant of the symmetric exchange axiom:

For all $F_1, F_2 \in \mathcal{F}$ there exist $a, b \in F_1 \triangle F_2$ such that $F_1 \triangle \{a, b\}, F_2 \triangle \{a, b\} \in \mathcal{F}$.

Unlike for matroids (where many apparently stronger exchange axioms define the same object) this is not equivalent to the usual delta-matroid definition, but is more restrictive.

Question 3. 1. Give a delta-matroid that is not strong.

2. ★ Show that every linear delta-matroid is strong. Hint: Reduce to the case that $D = D(A)$, $F_1 = \emptyset$ and $F_2 = V$. Argue that the fundamental graphs of A and $A * V$ have a common edge.

In fact, every even delta-matroid is strong. Sadly the proof of this is a little bit too messy to fit in an exercise, but we can use it as a fact to investigate the notion further.

Question 4. Show that the following are equivalent.

1. D is a strong delta-matroid
2. D is the projection of an even delta-matroid
3. D is the elementary projection of an even delta-matroid

Note: An elementary projection is a projection where the set you project away has cardinality one.